

Open-source SpaceWire CODEC IP Core User Manual

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Contents

1. Overview

2. License and Disclaimer

3. Block Diagram

4. Module Functions

4.1. TRANSMITTER

4.2. RECEIVER

4.3. STATEMACHINE

4.4. TIMER

4.5. TIMECODE

4.6. STATISTICAL INFORMATION

4.7. TRANSMITFIFO

4.8. RECEIVEFIFO

5. Interface Signals

6. Timing Diagrams

6.1. TRANSMITFIFO

6.2. RECEIVEFIFO

6.3. Transmit Time Code

6.4. Receive Time Code

7. Package

8. File Structure

9. Performance

10. Timing Constraints

10.1. UCF File

10.2. Clock

1

1

2

3

3

3

4

5

5

5

5

5

6

9

9

9

10

10

11

12

13

14

14

15

1. Overview

SpaceWire is a communication interface and communication protocol specification for data communication between onboard components of spacecraft (satellites).

The SpaceWire CODEC IP is a VHDL core intended to implement SpaceWire on Xilinx and Altera FPGAs.

It is designed to comply with ECSS-E-ST-50-12C and supports communication rates of up to 100 Mbps.

By incorporating it into an RMAP Target IP or SpaceWire Router IP, telemetry-data collection and SpaceWire router functions can be implemented.

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3. Block Diagram

The SpaceWire CODEC IP block diagram is shown below.

The Receiver decodes input from SpaceWire and writes it to the ReceiveFIFO.

The Transmitter converts data written to the TransmitFIFO into Data and Strobe signals and outputs them to SpaceWire.

Block / Signal	Description
Transmit Data	Data transmitted from the CODEC to SpaceWire
Receive Data	Data received from SpaceWire by the CODEC
Dout / Sout	SpaceWire Data and Strobe outputs
Din / Sin	SpaceWire Data and Strobe inputs
Link Control	Link-control signals
Link Status	Link-status information
Receive Time Code	Received time-code interface
Transmit Time Code	Transmit time-code interface
CreditCount	Remote-side transmit credit count
Statistical Information	Counters for link and character events
Receiver / Receive FIFO	Receive path and receive buffer
Time Code	Time-code processing
State Machine	Link initialization and operating-state control
Timer	Link timing generation
Transmitter / Transmit FIFO	Transmit path and transmit buffer

4. Module Functions

4.1. Transmitter

Only when transmission is enabled by the StateMachine, the Transmitter converts the data stored in the TransmitFIFO into Data and Strobe signals and outputs N-Chars corresponding to the received FCT count to SpaceWire. When there is no data to transmit or the number of transmittable characters is zero, NULLs are output.

The Transmitter manages the number of FCTs transmitted and received.

Each time the Receiver receives an FCT, the Transmitter adds 8 to the number of characters that can be transmitted. When the Transmitter sends an N-Char, the number of transmittable characters is decremented by 1.

When the Transmitter sends an FCT, the number of characters that can be received is increased by 8. When the Receiver receives an N-Char, the number of receivable characters is decremented by 1.

If the transmittable-character count becomes 56 or more, or the receivable-character count is 0 when the Receiver receives an N-Char, the Receiver outputs a credit error to the StateMachine.

The transmission rate is determined by TransmitClock and transmitClockDivideValue.

The transmission rate can be changed even while LinkUP is active by changing transmitClockDivideValue. During link initialization, operation is at 10 Mbps.

4.2. Receiver

The Data and Strobe signals are synchronized to the internal clock before being decoded.

The synchronization of the Data and Strobe signals is shown in the figure in the original manual.

When an N-Char (Data, EOP, or EEP) is received, the data is written to the ReceiveFIFO and the StateMachine is notified. The StateMachine is also notified when NULL, FCT, or TimeCode is received; FCT is additionally notified to the Transmitter.

Characters other than N-Char (Data, EOP, EEP) are not written to the ReceiveFIFO.

When an escape error or parity error is received, the data is discarded and the StateMachine is notified.

If the Data and Strobe signals do not change for 850 ns, this is reported to the StateMachine as a disconnect error.

Original figure labels: iSpaceWireSynchronize[1], iSpaceWireSynchronize[0], Din, Sin, ReceiveClock.

Receiver timing when NULL is received

The original manual provides a timing chart for the Receiver when a NULL is received.

4.3. StateMachine

The StateMachine manages link-interface initialization, normal operation, and error-recovery processing.

ErrorReset — EnableTransmitter and EnableReceiver are set to “L”, stopping the Transmitter and Receiver. After 6.4 μ s, the state changes to ErrorWait.

ErrorWait — EnableReceive becomes “H”, enabling the Receiver. After 12.8 μ s, the state changes to Ready. If the Receiver detects an error, the state changes to ErrorReset.

Ready — When LinkEnable becomes “H”, the state changes to Started. If the Receiver detects an error, the state changes to ErrorReset.

Started — EnableTransmit and SendNULLs become “H”, and the Transmitter sends NULLs. When the Receiver receives a NULL, the state changes to Connecting. If the Receiver detects an error, or 12.8 μ s elapses after entering Started, the state changes to ErrorReset.

Connecting — SendFCTs becomes “H”, and the Transmitter sends FCTs. When the Receiver receives an FCT, the state changes to Run. If the Receiver detects an error, or 12.8 μ s elapses after entering Connecting, the state changes to ErrorReset.

Run — SendTimeCode and SendNChar become “H”, allowing the Transmitter to send TimeCodes and N-Chars. If the Receiver detects an error, the state changes to ErrorReset.

Original timing-chart symbols: NULL, Sin, Din, ReceiveClock, iSpaceWireSynchronize; encoded bit sequence shown in the original figure: 10b, 11b, 01b, 11b, 10b, 11b, 10b, 00b.

4.4. Timer

Generates the 6.4 μ s and 12.8 μ s timers used during link initialization.

4.5. TimeCode

When the Receiver receives a TimeCode, it checks whether the received value is exactly one greater than the current value and outputs TickOut when it is valid.

When an invalid TimeCode is received, the current value is updated, but TickOut is not output.

4.6. Statistical Information

Counts EOP, EEP, and N-Char transmitted by the Transmitter; EOP, EEP, and N-Char received by the Receiver; and the number of LinkUP and LinkDown events during the link-up sequence.

Each item can count up to 4,294,967,296 events.

Setting statisticalInformationClear to "H" clears the accumulated counts.

4.7. TransmitFIFO

A 9×64 FIFO into which the host writes N-Chars that it wants to transmit.

A maximum of 56 N-Chars can be written. When 56 data items have been written, FULL becomes "H".

4.8. ReceiveFIFO

A 9×64 FIFO into which the Receiver writes decoded N-Chars (Data, EOP, EEP).

A maximum of 56 N-Chars can be stored. When no data has been written to the FIFO, Empty becomes "H".

5. Interface Signals

The SpaceWire CODEC IP operates on the rising edge of each Clock.

An input signal is enabled when it is set to “H”. An output signal indicates an active function when it is “H”.

The SpaceWire CODEC IP interface signals are as follows.

Signal	I/O	Description
Clock	I	System clock
transmitClock	I	Transmit clock
receiveClock	I	Receive clock
Reset	I	Reset
transmitFIFOWriteEnable	I	Write enable
transmitFIFODataIn[8:0]	I	Write data
transmitFIFOFull	O	FULL
transmitFIFODataCount[5:0]	O	Number of stored data items
receiveFIFOReadEnable	I	Read enable
receiveFIFODataOut[8:0]	O	Read data
receiveFIFOFull	O	FULL
receiveFIFOEmpty	O	Empty
receiveFIFODataCount[5:0]	O	Number of stored data items
tickIn	I	Transmit time code
timeIn[5:0]	I	Transmit time code
controlFlagsIn[1:0]	I	Transmit time-code control flags
tickOut	O	Received time code
timeOut[5:0]	O	Received time code
controlFlagsOut[1:0]	O	Received time-code control flags
linkStart	I	Set “H” to enable LinkStart. The link-up sequence automatically transitions from Ready to Started. Set “L” to disable LinkStart.
linkDisable	I	Set “H” to disable the link; set “L” to release Disable.
autoStart	I	Set “H” to enable AutoStart. The link starts when NULL is received; set “L” to disable AutoStart.
transmitClockDivideValue[5:0]	I	Transmission-rate setting after link-up. Transmission rate = TransmitClock[MHz] / (TransmitClockDivideValue + 1).
creditCount[5:0]	O	Remote-side credit count (number of characters that may be transmitted). Receiver adds 8 when an FCT is received; Transmitter subtracts 1 when an N-Char is transmitted.

Signal	I/O	Description
outstandingCount[5:0]	O	Local-node credit count (number of characters that may be received). Transmitter adds 8 when it sends an FCT; Receiver subtracts 1 when it receives an N-Char.
transmitActivity	O	Data being read
receiveActivity	O	Data being written
spaceWireDataIn	I	SpaceWire Data input
spaceWireStrobeIn	I	SpaceWire Strobe input
spaceWireDataOut	O	SpaceWire Data output
spaceWireStrobeOut	O	SpaceWire Strobe output
linkStatus[15:0]	O	Link status information
linkStatus[0]	—	EnableTransmit. Becomes “H” when link-up sequence enters Started; becomes “L” on link-down or an error during link-up.
linkStatus[1]	—	EnableReceive. Becomes “H” when link-up sequence enters ErrorWait; becomes “L” on link-down or an error during link-up.
linkStatus[2]	—	SendNULLs. Becomes “H” when link-up sequence enters Started; becomes “L” on link-down or an error during link-up.
linkStatus[3]	—	SendFCTs. Becomes “H” when link-up sequence enters Connecting; becomes “L” on link-down or an error during link-up.
linkStatus[4]	—	SendNChar. Becomes “H” when link-up sequence enters Run; becomes “L” on link-down or an error during link-up.
linkStatus[5]	—	SendTimeCode. Becomes “H” when link-up sequence enters Run; becomes “L” on link-down or an error during link-up.
linkStatus[6]	—	Reserved; fixed at “0”.
linkStatus[7]	—	SpaceWireReset. Becomes “H” when link-up sequence enters ErrorReset; becomes “L” when it enters ErrorWait.
linkStatus[8]	—	Link established indication. Becomes “H” when link-up sequence enters Run; becomes “L” when the link goes down.
linkStatus[9]	—	Becomes “H” for one clock when the Transmitter sends an N-Char.
linkStatus[10]	—	Becomes “H” for one clock when the Receiver receives an N-Char.
linkStatus[11]	—	Becomes “H” for one clock when the Receiver receives a NULL.
linkStatus[12]	—	Becomes “H” for one clock when the Receiver receives an FCT.
linkStatus[13]	—	Becomes “H” for one clock when the Receiver receives an EOP.
linkStatus[14]	—	Becomes “H” for one clock when the Receiver receives an EEP.
linkStatus[15]	—	Reserved; fixed at “0”.

Signal	I/O	Description
errorStatus[7:0]	O	Link error information
errorStatus[0]	—	CharacterSequenceError. Becomes "H" when an unexpected FCT or N-Char is received during the link-up sequence. Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[1]	—	CreditError. Becomes "H" when an N-Char is received while CreditCount is 56 or more, or OutstandingCount is 0. Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[2]	—	RxErr. Becomes "H" when a ParityError, DisconnectError, or ESCError is detected while EnableReceive is "H". Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[3]	—	Reserved; fixed at "0".
errorStatus[4]	—	Parity Error. Becomes "H" when a parity error is detected while EnableReceive is "H". Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[5]	—	DisconnectError. Becomes "H" when Data and Strobe do not change for 850 ns while EnableReceive is "H". Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[6]	—	ESCError. Becomes "H" when the control character received after an ESC character is not FCT while EnableReceive is "H". Becomes "L" when the link-up sequence enters ErrorReset.
errorStatus[7]	—	Reserved; fixed at "0".
statisticalInformationClear	I	Set "H" to clear statistical information. Set "L" to start counting.
statisticalInformation 8×[31:0]	O	Statistical information
0 [31:0]	—	Accumulated number of EOPs transmitted by the Transmitter
1 [31:0]	—	Accumulated number of EOPs received by the Receiver
2 [31:0]	—	Accumulated number of EEPs transmitted by the Transmitter
3 [31:0]	—	Accumulated number of EEPs received by the Receiver
4 [31:0]	—	Accumulated number of N-Chars transmitted by the Transmitter
5 [31:0]	—	Accumulated number of N-Chars received by the Receiver
6 [31:0]	—	Accumulated number of times the link-up sequence established the link
7 [31:0]	—	Accumulated number of times the link-up sequence brought the link down

6. Timing Diagrams

The timing charts for the SpaceWire CODEC IP interface signals are shown in this section.

6.1. TransmitFIFO

This is the FIFO into which the host writes N-Chars to be transmitted.

During a TransmitFIFO write, if WriteEnable is “H” at the rising edge of Clock, data can be written to the FIFO.

When 56 data items have been written, FIFOFull becomes “H”.

Note: Even when FIFOFull is “H”, up to 7 additional data items can subsequently be written. However, if 8 or more additional data items are written, DataCount will become misaligned and data can no longer be written correctly. Take care when using the FIFO in this condition.

6.2. ReceiveFIFO

This FIFO contains N-Chars received over SpaceWire.

Control characters other than EOP and EEP are not written to the ReceiveFIFO.

During a ReceiveFIFO read, if ReadEnable is “H” at the rising edge of Clock, data can be read from the FIFO.

Note: When FIFOEmpty is “H”, data cannot be read. Take care when performing continuous reads.

Original timing-chart labels: Clock; receiveFIFOReadEnable; receiveFIFOEmpty; receiveFIFODataOut[8:0]; receiveFIFODataCount[5:0]; transmitFIFOWriteEnable; transmitFIFOFull; transmitFIFODataIn[8:0]; transmitFIFODataCount[5:0].

The original figures show hexadecimal/count values including 06h, 05h, 01h, 00h, 33h–38h, 03h, 02h, 034h–038h, and “xxx”.

6.3. Transmit Time Code

The timing chart for transmitting a TimeCode is shown in the original manual.

When the clock-synchronous TickIn signal is asserted, the values of TimeIn and ControlFlagsIn are transmitted as a TimeCode.

Update the TimeIn and ControlFlagsIn values for each TickIn.

The Transmitter ignores TickIn signals while a TimeCode is being transmitted.

The TickIn period must have an interval of at least 25 transmit-rate periods.

6.4. Receive Time Code

The timing chart for receiving a TimeCode is shown in the original manual.

When a TimeCode is received, a Clock-synchronized TickOut signal is output.

The received TimeCode value is output on TimeOut and ControlFlagsOut.

Original figure labels include: Clock, tickOut, controlFlagsOut[1:0], timeOut[5:0], tickIn, controlFlagsIn[1:0], timeIn[5:0], P, TIMECODE, T0–T7, NULL, and the validity sequence “Valid / Invalid / Invalid / Valid”.

7. Package

The Constant declarations defined in the package must be changed according to the input clock. The Constant declarations are shown below.

Constant name	Type / bit width	Description
gDisconnectCountValue	integer, 0 to 255	Disconnect-error time setting. A disconnect error is generated if the DS signal does not change for approximately 850 ns. Setting value = $850 / T$. Default: 141 (166 MHz).
gTimer6p4usValue	integer, 0 to 1023	Setting for generation time of the after6p4us signal required for StateMachine transitions. Generates after6p4us at 6.4 μ s. Setting value = $6400 / T$. Default: 320 (50 MHz).
gTimer12p8usValue	integer, 0 to 2047	Setting for generation time of the after12p8us signal required for StateMachine transitions. Generates after12p8us at 12.8 μ s. Setting value = $12800 / T$. Default: 640 (50 MHz).
gInitializeTransmitClockDivideValue	std_logic_vector(5 downto 0)	Transmit-rate setting during initialization. Configured so that the initial transmission rate is 10 Mbps. Setting value = $1 / (10 / f[\text{MHz}]) - 1$. Default: 9 (100 MHz).

Where: f = TransmitClock frequency; T = ReceiveClock period for the relevant timer; T = Clock period where applicable. The original document labels the timing variable as T in the formulas.

8. File Structure

The SpaceWire CODEC IP file structure is shown below.

File name	Description
SpaceWireCODECIP.vhdl	SpaceWire CODEC IP top-level module
SpaceWireCODECIPReceiverSynchronize.vhdl	SpaceWire receive module
SpaceWireCODECIPTransmitter.vhdl	SpaceWire transmit module
SpaceWireCODECIPStateMachine.vhdl	Link-interface state-machine module
SpaceWireCODECIPTimer.vhdl	Timer module
SpaceWireCODECIPTimeCodeControl.vhdl	TimeCode receive module
SpaceWireCODECIPStatisticalInformationCount.vhdl	Statistical-information counter module
SpaceWireCODECIPFIFO9x64.vhdl	FIFO RTL module
SpaceWireCODECIPLinkInterface.vhdl	Link-interface top-level module
SpaceWireCODECIPSynchronizeOnePulse.vhdl	Clock-synchronization module
SpaceWireCODECIPPackage.vhdl	Package containing Constant declarations

Note: The character encoding of the above source files is UTF-8.

9. Performance

The resource utilization of the SpaceWire CODEC IP is shown below.

Device: Xilinx Spartan-6 XC6SLX75

Synthesis tool: ISE 14.4

Resource utilization varies depending on synthesis options and device.

Module	Flip-Flops	LUT	Slices	RAM (in LUT)
SpaceWireCODECIP	691	662	306	24
SpaceWireCODECIPReceiverSynchronize	59	104	39	0
SpaceWireCODECIPTransmitter	114	116	63	0
SpaceWireCODECIPStateMachine	28	30	27	0
SpaceWireCODECIPTimer	24	25	24	0
SpaceWireCODECIPTimeCodeControl	25	9	9	0
SpaceWireCODECIPStatisticalInformationCount	274	266	86	0
SpaceWireCODECIPFIFO9x64	82	77	25	12

10. Timing Constraints

10.1. UCF File

Timing constraints are required to ensure correct timing of the synthesized circuit.

Timing constraints are required for all clock-domain crossings.

The timing constraints for a communication rate of 100 Mbps are as follows:

clock = 50 MHz
transmitClock = 100 MHz
receiveClock = 166.6 MHz

```
NET "clock" TNM_NET = "SYS_CLK";<br/>NET "transmitClock" TNM_NET = "TX_CLK";<br/>NET "receiveClock" TNM_NET =  
"RX_CLK";<br/>TIMESPEC TS_SYS_CLK_to_TX_CLK = FROM "SYS_CLK" TO "TX_CLK" TIG;<br/>TIMESPEC TS_TX_CLK_to_SYS_CLK  
= FROM "TX_CLK" TO "SYS_CLK" TIG;<br/>TIMESPEC TS_TX_CLK_to_RX_CLK = FROM "TX_CLK" TO "RX_CLK"  
TIG;<br/>TIMESPEC TS_RX_CLK_to_TX_CLK = FROM "RX_CLK" TO "TX_CLK" TIG;<br/>TIMESPEC TS_SYS_CLK_to_RX_CLK = FROM  
"SYS_CLK" TO "RX_CLK" TIG;<br/>TIMESPEC TS_RX_CLK_to_SYS_CLK = FROM "RX_CLK" TO "SYS_CLK" TIG;<br/>TIMESPEC  
"TS_clk" = PERIOD "SYS_CLK" 20 ns HIGH 50 %;<br/>TIMESPEC "TS_txclk" = PERIOD "TX_CLK" 10 ns HIGH 50  
%;<br/>TIMESPEC "TS_rxclk" = PERIOD "RX_CLK" 6 ns HIGH 50 %;
```

10.2. Clock

The SpaceWire CODEC IP must be supplied with Clock, TransmitClock, and ReceiveClock.

The recommended clocks for a communication rate of 100 Mbps are:

Parameter	Recommended value
Clock	50–100 MHz
TransmitClock	100.0 MHz
ReceiveClock	166.6 MHz
gInitializeTransmitClockDivideValue	001001
gDisconnectCountValue	141

Input-clock conditions:

- ReceiveClock should be approximately 1.5 times the receive rate.
- When TransmitClock is used below 100 MHz, reduce ReceiveClock by the same ratio.
- If ReceiveClock is not reduced by the same ratio when TransmitClock is reduced, there are combinations of transmit/receive rates for which consecutive FCT control codes may not be processed correctly.

Example 1: When TransmitClock is used below 100 MHz

Parameter	Value
Clock	50.0 MHz
TransmitClock	50.0 MHz
ReceiveClock	83.3 MHz
gInitializeTransmitClockDivideValue	000100
gDisconnectCountValue	71

Example 2: A clock combination that cannot correctly process consecutive FCT reception

Parameter	Value
Clock	50.0 MHz
TransmitClock	10.0 MHz
ReceiveClock	166.6 MHz
gInitializeTransmitClockDivideValue	000000
gDisconnectCountValue	141

If the example-2 input clocks are used with a 10 Mbps transmit rate and 100 Mbps receive rate, and FCTs are received consecutively within the 100 ns TransmitClock period, two or more received FCTs may nevertheless be processed as only one FCT.

If this condition continues, the credit count may ultimately fail to increase beyond 8.

End of English translation.